Chemical Engineering Journal 193–194 (2012) 215–226

Contents lists available at SciVerse ScienceDirect

Chemical Engineering Journal

journal homepage: www.elsevier.com/locate/cej

Synthesis of monodispersed polyisoprene–silica nanoparticles via differential microemulsion polymerization and mechanical properties of polyisoprene nanocomposite

Anong Kongsinlark [a], Garry L. Rempel [b,*], Pattarapan Prasassarakich [c,d,*]

[a] Program in Petrochemistry and Polymer Science, Faculty of Science, Chulalongkorn University, Bangkok 10330, Thailand
[b] Department of Chemical Engineering, University of Waterloo, ON, Canada N2L3G1
[c] Department of Chemical Technology, Faculty of Science, Chulalongkorn University, Bangkok 10330, Thailand
[d] Center for Excellence on Petrochemical and Materials Technology, Chulalongkorn University, Bangkok 10330, Thailand

ARTICLE INFO

Article history:
Received 17 January 2012
Received in revised form 27 March 2012
Accepted 2 April 2012
Available online 9 April 2012

Keywords:
Monodispersion
Polyisoprene
Nanoparticle
Nanosilica
Nanocomposite

ABSTRACT

Monodispersed polyisoprene–SiO_2 nanoparticles were successfully synthesized by a differential microemulsion polymerization of isoprene on silane treated nanosilica. Core–shell morphology was formed consisting of silica as the nano-core encapsulated by polyisoprene (PIP) as the nano-shell, as confirmed by TEM. PIP–SiO_2 nanoparticles were produced at 20–60 nm with a narrow size distribution resulting in a reduced nano-SiO_2 aggregation in the PIP matrix. The influence of surfactant concentration, monomer/water ratio and SiO_2 loading on particle size, monomer conversion as well as grafting efficiency was investigated. The monomer conversion was 87% and the polymer grafting efficiency was as high as 78% at an extremely low surfactant concentration (3 wt.% based on monomer). For rubber applications, the PIP–SiO_2 nanocomposite has been used as an effective nano-filler in natural rubber (NR) latex and the NR filled with PIP–SiO_2 prevulcanizate clearly showed an improvement in the storage modulus, tensile strength, tensile modulus, and anti-ageing properties.

© 2012 Elsevier B.V. All rights reserved.

1. Introduction

Nanocomposite materials have increasing potential for future applications due to their extraordinary properties based on the combined system of inorganic species with organic polymers. The composite particles are designed to enhance the overall performance with respect to mechanical, electrical and thermal properties compared to pure polymers [1–3]. However, these remarkable properties could not be realized until recently due to the different polarity nature between filler (hydrophilic) and polymer (hydrophobic) resulting in poor dispersion, inferior compatibility and less stability of the filler in the polymer matrix. A critical challenge in the design of these composites is the control of the mixing between the two dissimilar phases, and hence many chemical methods have been developed for the preparation of polymer nanocomposites. Polystyrene/silica encapsulation using dispersion polymerization in aqueous ethanol medium with poly(*N*-vinyl pyrrolidone) as stabilizer was earlier reported and a nanocomposite particle size of 0.2–0.5 μm was obtained [4]. The technique of suspension–dispersion polymerization was also applied to prepare poly(styrene-butylacrylate-acrylic acid)-graft–silica (PSBA-g-silica) hybrid nanoparticles resulting in an improvement in the UV-shielding and water resistance properties [5].

A modified emulsion polymerization of microemulsion or miniemulsion polymerization has been developed to obtain the nanosized particles in an aqueous system. For the preparation of polymethylmethacrylate (PMMA)/carbon nanotube composites by microemulsion polymerization, uniform composite film could be obtained and the electrical properties could also be enhanced, compared with the composites from solution mixing of PMMA and carbon nanotubes [6]. In addition, polystyrene/silica was synthesized using miniemulsion polymerization with a particle size of 218 nm at a surfactant level of 14 wt.% based on monomer [7]. Later, PMMA/SiO_2 nanocomposite particles could be prepared through microemulsion polymerization by using silica particles coated with 3-(trimethoxysilyl) propyl- methacrylate (MSMA) and a particle size of 80 nm was obtained when using a high surfactant amount of about 40 wt.% based on monomer [8].

However, both microemulsion and miniemulsion polymerization require a high amount of surfactant. The surfactants are not only expensive but also have a significant negative impact on the physical properties of the resulting polymers. Moreover, the concentration of such surfactant molecules in solution needs to be tuned carefully because a latex formation in the free micelles in

* Corresponding authors. Tel.: +1 519 8884567x32702; fax: +1 519 7464979.
E-mail addresses: grempel@uwaterloo.ca (G.L. Rempel), ppattara@chula.ac.th (P. Prasassarakich).

1385-8947/$ - see front matter © 2012 Elsevier B.V. All rights reserved.
http://dx.doi.org/10.1016/j.cej.2012.04.008

the emulsion is observed at higher concentrations. These molecules are usually only weakly bonded to the silica surface and can therefore desorb easily [9]. Since, it is desirable to reduce the surfactant amount under mild reaction conditions and also to yield nanosized particles with high conversion, the differential microemulsion polymerization method was developed. Differential microemulsion polymerization has been developed in which the monomer feed was provided continuously as very small droplets and slowly added to the polymerization system under mild agitation. Compared with conventional emulsion polymerization, the differential microemulsion polymerization requires an extremely low surfactant concentration and is controllable in providing PMMA nanoparticles of less than 20 nm at a surfactant concentration of 5% based on monomer weight, which is difficult to be realized by conventional methods [10].

This method was confirmed to have high potential for the synthesis of a nanosized PMMA using 2,2′-azobisisobutyronitrile (AIBN) as oil soluble initiator and particles of less than 50 nm were produced using 5 g/L surfactant concentration [11]. The glycidyl-functionalized PMMA with a particle size around 50 nm consisting of a nanosized core of PMMA and a nanosized thin shell of the poly(MMA-co-glycidyl methacrylate) was successfully synthesized by differential microemulsion polymerization using a small amount of surfactant of 1/217 based on monomer [12]. Polystyrene encapsulated silica synthesized by differential microemulsion polymerization was used as filler in NR [13] and in carboxylated styrene butadiene rubber (XSBR)/NR [14]. The tensile properties, dynamic mechanical properties, and thermal stability were all improved by the addition of PS-SiO_2 within the rubber.

Interestingly, a differential microemulsion polymerization is advantageous for the production of nanosized polymer latex with an extremely low surfactant concentration and is normally performed using an environmental friendly process due to the absence of organic solvents under mild conditions. For the mechanism of differential microemulsion polymerization, when the reaction was initiated, the system contains initiators, radicals of primary radicals, monomer radicals, and polymer radicals, nonaggregated surfactants, and a very small amount of monomer [15]. The initiator decomposed in the aqueous phase to generate primary radicals and some of the radicals entered into monomer swollen micelles to produce polymer particles. So far, the synthesis of nanosized rubber-SiO_2 by differential microemulsion polymerization has not been reported.

Many previous works have reported on the improvement of the mechanical properties of NR filled with SiO_2 composites [16–18]. However, difficulties of such an approach for NR/silica composites are due to potential incompatibilities between silica and NR due to the difference of polarity and surface energy, resulting in a phase separation, filler agglomeration and the weakening of mechanical and thermal properties. Admicellar polymerization [19] and radiation induced admicellar polymerization [20] have been used to prepare polyisoprene/silica composites with superior mechanical properties compared to *cis*-1,4 polyisoprene rubber/unmodified SiO_2. In this work, a differential microemulsion polymerization was applied to synthesize the monodispersed PIP–SiO_2 nanocomposites with a uniform particle size distribution and PIP–SiO_2 nanoparticles have been used as effective fillers in NR.

In this research work, the monodispersion of nanosized polyisoprene–SiO_2 composite with core/shell morphology was firstly synthesized via differential microemulsion polymerization and was used as an effective nano-filler in polyisoprene for a new class of polyisoprene–SiO_2 nanocomposites. The nano-SiO_2 was pretreated using organo silane coupling agents to produce a siloxane bond at the silica surface and the carbon double bonds at the chain end of the silane coupling agent could be polymerized with the isoprene monomer. The influence of initiator and surfactant concentration, monomer/water ratio and silica loading on particle size, monomer conversion as well as encapsulation efficiency was also investigated. The stress–strain behavior and thermal ageing properties of polyisoprene–SiO_2 nanocomposites were also studied.

2. Experimental

2.1. Materials

Nano-SiO_2 (Aerosil 200) with an average particle size of 12 nm was supplied by Degussa (Thailand). The 3-methylacryloxypropyl trimethoxy silane (MPS, Sigma) and vinyl trimethoxysilane (VTS, Sigma) were used as coupling agents. An ammonia solution (25% NH_4OH, BDH) was used as catalyst for silica surface modification. Isoprene monomer (IP, Aldrich), sodium dodecyl sulfate (SDS, Fisher Scientific), sodium persulfate (SPS, Aldrich) and sodium bicarbonate ($NaHCO_3$, VWR) were used for synthesis of the polyisoprene–SiO_2 emulsion. The methanol (MeOH, Fisher Scientific) and methyl ethyl ketone (MEK, Fisher Scientific) were used for the modified silica precipitation and the rubber coagulation, respectively. De-ionized water was also used in all experiments.

For the preparation of prevulcanized NR nanocomposites, high ammonia natural rubber latex (NRL) with 60% dry rubber content, sulfur as a vulcanizing agent, zinc oxide (ZnO) and zincdiethyl dithiocarbamate (ZDEC) as vulcanization accelerators were obtained from the Rubber Research Institute of Thailand.

2.2. Pretreatment of nano-SiO_2

10 g of nanosilica was initially dispersed in 300 mL of de-ionized water with stirring at 450 rpm for 30 min. Then, 0.3 g of the coupling agent (MPS, VTS) was added dropwise into the dispersion and the pH of the solution was then adjusted to around 10 with the addition of 25 wt.% aqueous ammonia solution. The dispersion was stirred for 30 min at room temperature, and then heated up to a reaction temperature of 80 °C with continuous stirring overnight. Afterwards, the suspension was dried at 90 °C for 24 h to obtain a modified-SiO_2. A Soxhlet extraction was used to remove coupling agents which were not effectively bonded on the silica surface. Finally, the nano-SiO_2 powders (MPS–SiO_2, VTS–SiO_2) produced were dried in an oven at 55 °C for 48 h.

2.3. Synthesis of polyisoprene–silica nanocomposite

Nanosized PIP–SiO_2 particles were prepared in a 250 mL Parr reactor equipped with a thermocouple, a feeding tube, a coolant, and a magnetic stirrer under a nitrogen atmosphere. Typically, 2 g of modified nano-SiO_2 (VTS–SiO_2) was dispersed in 20 mL of de-ionized water with the aid of an ultrasonic bath for 1 h. Then, 0.2 g of SPS initiator, 0.6 g of SDS surfactant and 0.7 g of $NaHCO_3$ was dissolved in 50 mL of distilled water. Then, the solution and the modified nano-SiO_2 was charged into the Parr reactor and stirred at 300 rpm for 45 min. After degassing, the solution was heated up to the reaction temperature of 75 °C. For differential microemulsion polymerization, 20 g of the condensed isoprene was slowly fed dropwise into the reactor using a peristaltic pump at a feed rate of 0.4 mL/min. When the addition of the monomers was complete, the solution was stirred at a specific temperature for an additional 18 h. Then, the system was cooled down to room temperature and the composite latex was precipitated with an excess of methyl ethyl ketone to produce the coagulated rubber. The coagulated rubber was dried at 50 °C overnight. Furthermore, the coagulated rubber composite was then extracted using petroleum ether in a Soxhlet apparatus to remove the free polymer (ungrafted polyisoprene) and then dried to a constant weight.

The monomer conversion was determined by a gravimetric method and calculated using Eq. (1):

$$\text{Conversion}\,(\%) = (M_0 - M_1)/M_2 \times 100\% \quad (1)$$

where M_0 is the mass of the resulting composite particles, M_1 is the mass of the charged SiO_2 particles, and M_2 is the mass of the charged monomer.

The polymer grafting efficiency (%GE) was calculated using Eq. (2):

$$\text{Grafting efficiency} = \frac{\text{Amount of polymer in the composite sample}}{\text{Total amount of polymer formed}} \times 100 \quad (2)$$

The silica encapsulation efficiency was measured by an acid etching method as shown in Scheme 1 [21]. The free silica particles were separated from the PIP–SiO_2 latex by HF etching. Trial experiments were conducted in which a HF solution was added to bare silica particles, PIP and PIP–SiO_2 latex. After HF etching, it was found that all of the bare silica particles were consumed whereas the weight of PIP particles remained constant and the PIP–SiO_2 only lost the weight of the free silica particles and the incompletely encapsulated SiO_2. Therefore, completely encapsulated SiO_2 was not extracted by HF etching. The weight percent of residue (encapsulated SiO_2) was measured gravimetrically.

2.4. Characterization

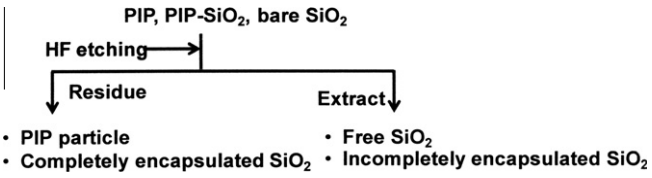
Modified-silica and PIP–SiO_2 were characterized using Fourier transform infrared spectroscopy (Bruker 3000X spectrometer). Before measurement, all samples were purified via soxhlet extraction to remove the unreacted coupling agent and ungrafted polyisoprene, the grafted PIP–SiO_2 samples were ground with dried KBr powder and compressed into a pellet.

The number-average diameter (D_n) and size distribution of PIP–SiO_2 nanoparticles were measured using a Dynamic Light Scattering technique (Malvern Instrument, USA). The morphology of PIP–SiO_2 nanoparticle was examined by a transmission electron microscope (TEM) (JEOL, 80 kV). The samples diluted 20 times were stained with OsO_4 and dropped on a copper grid.

The amount of silica content in the PIP–SiO_2 was determined using thermogravimetric/differential thermal analysis (TG/DTA) on a Perkin–Elmer Pyris Diamond instrument. The samples were placed into a platinum pan. The temperature was raised under a nitrogen atmosphere from room temperature to 700 °C at a constant heating rate of 10 °C/min. The nitrogen gas flow rate was 50 mL/min.

2.5. Preparation of NR nanocomposite

For the preparation of prevulcanized polyisoprene nanocomposite, natural rubber latex (NRL) with a total solid content of 60% was selected to blend with PIP–VTS–SiO_2 or PIP–MPS–SiO_2. The latices of PIP–VTS–SiO_2 and PIP–MPS–SiO_2 were dropped into NRL at various weight ratios (NR: PIP–VTS–SiO_2 and NR: PIP–MPS–SiO_2 = 100:0, 95:5, 90:10, 80:20, 70:30) under a stirring rate of 450 rpm for 30 min to form a good dispersion of NR/PIP–SiO_2 composite. Then, sulfur (1.5 phr) as vulcanizing agent, ZDEC (1 phr) and ZnO (2 phr) as accelerators were added into the mixture and the temperature was raised to 60 °C with constant stirring at 300 rpm for 2 h. Then, the NR nanocomposite latex was cooled to room temperature and cast on a raised glass plate having a dimension of 13 cm × 13 cm × 3 mm. The cast sheet was dried at 70 °C for 5 h. For thermal ageing of composites, tensile specimens were aged at 100 °C for 24 h in an air circulating ageing oven and the mechanical properties of the samples before and after ageing were measured to estimate the ageing resistance.



Scheme 1. Schematic diagram of HF etching method.

2.6. Mechanical properties of NR nanocomposites

Dynamic mechanical thermal analysis of PIP–SiO_2 and NR/PIP–SiO_2 nanocomposite was performed using a dynamic mechanical thermal analyser (DMA, METTLER). NR nanocomposites were measured with a strain of 70 μm peak to peak displacement over the temperature range of -70 °C to 60 °C at a frequency of 10 Hz and the heating rate was 5 °C/min. The storage modulus and the loss tangent ($\tan\delta$) curve dependence on temperature was observed.

Mechanical properties in terms of tensile strength, %elongation at break and modulus were measured by a Universal Testing Machine (LLOYD model LR5K), according to the ASTM-412 method at 500 mm/min of the cross-head speed. All samples were cut into dumbbell-type specimens using a Wallance die cutter. The data points were averaged from three measurements of the four specimens, and stress–strain curves were also recorded.

3. Results and discussion

3.1. Characterization of polyisoprene–SiO_2 nanocomposites

From Fig. 1a, the spectrum of SiO_2-treated with MPS (MPS–SiO_2), the peaks at 2960, 1732 and 1633 cm^{-1} are attributed to =CH_2, C=O and C=C stretching of the MPS groups, respectively [22–24]. The peak at 3441 cm^{-1} is assigned to the surface hydroxyl group (O–H) of silica [13,25]. While a very broad band appearing at 1111 cm^{-1}, together with two less pronounced bands at 810 and 473 cm^{-1} correspond to the vibration absorption band of the Si–O–Si groups. For silica modified with VTS (VTS–SiO_2) (Fig. 1b), the absorption bands at 1114, 802 and 470 cm^{-1} correspond to Si–O–Si groups. The peaks at 3443, 3067 and 2962 cm^{-1} relate to OH, CH and CH_2 stretching of the VTS groups. The bands at 1604 cm^{-1} (C=C) and 1411 cm^{-1} (CH out of plain bending) are attributed to the double bonds of VTS. These results indicate that coupling agents could be bonded onto the silica surface. The methoxy groups of silane were firstly hydrolysed by water in the aqueous solution to form silanol groups. The added ammonium hydroxide catalysed the hydrolysis reaction due to a nucleophilic attack at the silicon atom of the silane. The siloxane linkage between the silica surface and the silane agent was then produced through a polycondensation reaction [26].

For the PIP–VTS–SiO_2 composite (Fig. 1c), new peaks at 2860 and 2972 cm^{-1} are related to the methyl and methylene stretching of polyisoprene, respectively. Additionally, the low intensity band at 837 cm^{-1} and medium intensity band at 1674 cm^{-1} correspond to the CH wag of trisubstituted olefin and the C=C stretching of polyisoprene, respectively. The methyl deformation bands of polyisoprene were apparent at 1370 and 1490 cm^{-1} [27]. These results imply that polyisoprene was grafted onto the silica surface.

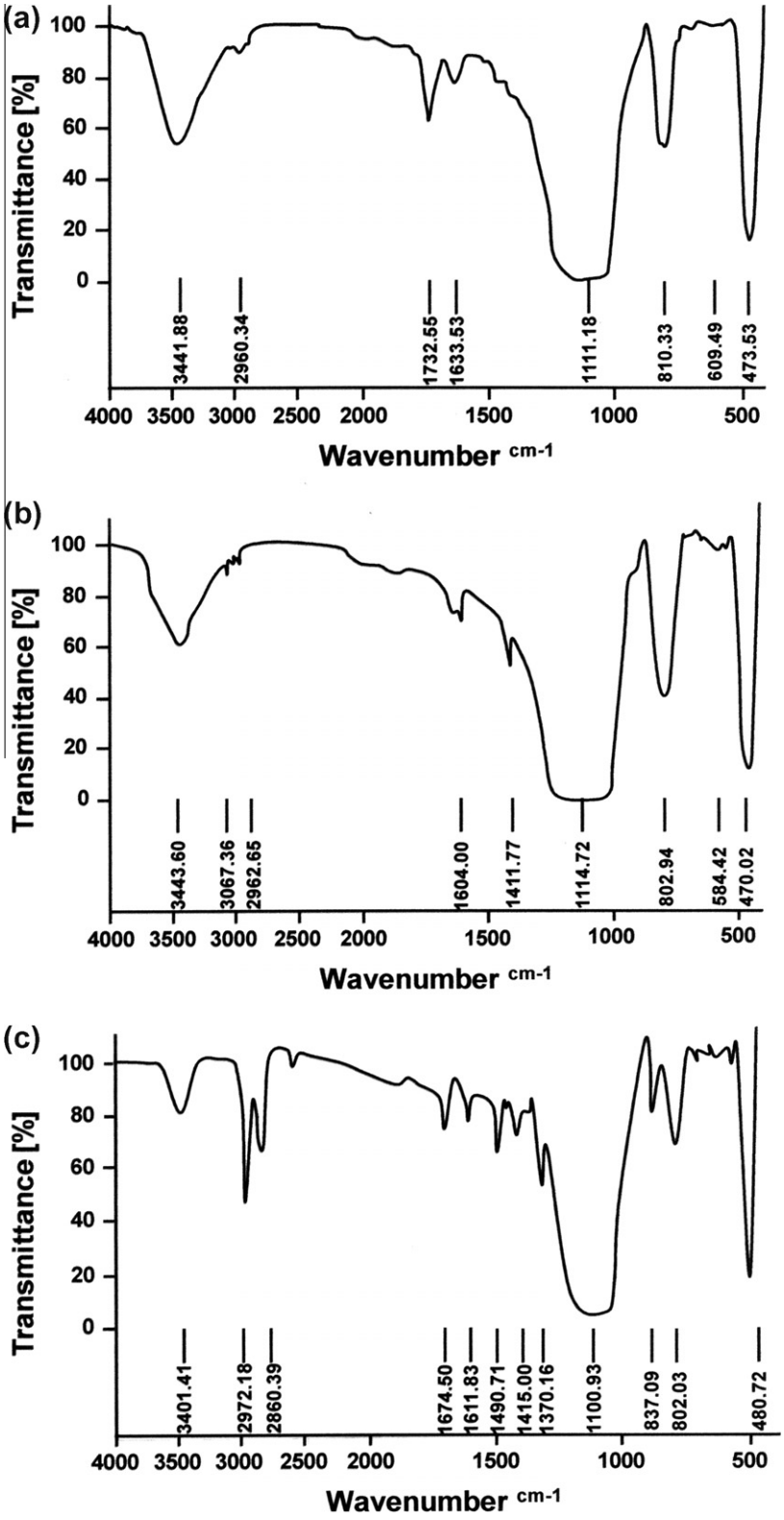


Fig. 1. FTIR spectra of (a) MPS–SiO_2, (b) VTS–SiO_2, (c) PIP–VTS–SiO_2.

3.2. Encapsulation of VTS–SiO_2 with polyisoprene

The encapsulation of the nanosized VTS–SiO_2 with PIP was studied under various process parameters such as monomer to water ratio, surfactant concentration, initiator amount and silica loading. The effects of parameters on particle size, polymer grafting efficiency, monomer conversion as well as silica encapsulation efficiency are discussed in detail.

3.2.1. Influence of monomer/water and initiator concentration

The monomer to water ratio significantly affected the encapsulation as shown in Fig. 2. The particle size of PIP–SiO_2

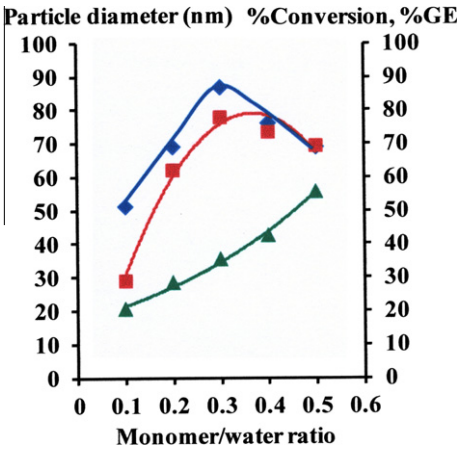


Fig. 2. Effect of monomer to water ratio (M/H_2O) on PIP–SiO_2; (▲) Particle diameter, (♦) %Conversion, (■) %GE. Condition: SPS = 1 wt.%, SDS = 3 wt.%, VTS–SiO_2 = 10 wt.% based on monomer.

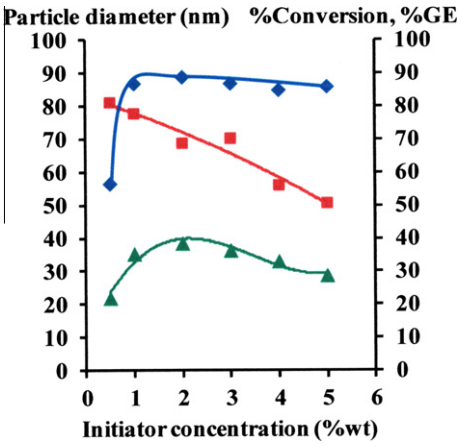


Fig. 3. Effect of initiator concentration on PIP–SiO_2; (▲) Particle diameter, (♦) %Conversion, (■) %GE. Condition: M/H_2O = 0.3, SDS = 3 wt.%, VTS–SiO_2 = 10 wt.% based on monomer.

nanocomposite increased with increasing monomer concentration. The particle diameter was increased from 20 nm to 55 nm when the monomer to water ratio was increased from 0.1 to 0.5. At a high monomer to water ratio, the probability that one particle collided with another particle was higher and thus monomer conversion increased which resulted in a larger particle size than that obtained at low monomer concentration [10]. The monomer conversion and grafting efficiency were increased when increasing the monomer to water ratio from 0.1 to 0.3 and then decreased at a higher ratio. Thus, a high monomer to water ratio above 0.3 caused a negative effect on particle size, conversion and polymer grafting efficiency. However, at the highest monomer to water ratio (0.5), most of the further added monomer molecules had to enter into the existing polymer particles, which were more than those of the newly nucleated particles and the long time needed for monomer diffusion into the particle would decrease the monomer conversion [28]. Thus, the monomer conversion was decreased at high monomer/water ratio. The grafting efficiency (GE) had a similar trend to conversion and the %GE at higher monomer concentration was decreased, due to the fact that the free PIP was more pronounced than grafted PIP [29]. The maximum conversion (87%), grafting efficiency (78%) and small particle size (35 nm) were obtained at a monomer to water ratio of 0.3.

The effect of initiator concentration on SiO_2 encapsulation with PIP was also studied over the range of 0.5–5 wt.% based on monomer. From Fig. 3, it is seen that the polymer grafting efficiency was dramatically affected by initiator amount. The %GE decreased from 80% to 50% with an increase in initiator concentration from 0.5 to 5 wt.%. It can be explained in that, a large amount of oligoradicals are generated in the aqueous phase, but these radicals can enter into the polymer particles to react with the modified silica double bonds. A greater radical entry rate may be the reason for the decrease in the grafting efficiency with the increase in initiator concentration. This phenomenon resulted in a higher free PIP concentration rather than the grafting of PIP onto the silica surface. Although the high initiator amount had a negative effect on the grafting efficiency, the increasing initiator level had a positive effect on the monomer conversion. At high initiator concentration, the initiator decomposed to generate free radicals and the radicals reacted with monomer to produce primary radicals and then the growing polymer chains increased resulting in an increase in conversion [30,31]. Therefore, a low initiator concentration was more favorable for PIP/SiO_2 encapsulation.

3.2.2. Influence of surfactant concentration and silica loading

SDS was used as an anionic surfactant for encapsulation of nano-SiO_2 with PIP. The amount of surfactant had a great influence on particle size, conversion and polymer grafting efficiency as illustrated in Fig. 4. Particle size decreased from 50 nm to 20 nm with an increase in the surfactant amount from 0.5 to 10 wt.% based on monomer. At a high surfactant level, smaller nuclei in the system were produced and then a smaller particle size of PIP–SiO_2 composite was formed [32]. The conversion was increased from 60% to 90% with an increase in surfactant level from 0.5% to 7% it then levelled off. An increase in the SDS amount increased the number of monomer-swollen micelles and therefore the number of particles which led to an increase in the polymerization rate and the final conversion. For PIP–SiO_2 encapsulation, polymer grafting efficiency was significantly affected by the surfactant concentration; %GE was increased with increasing SDS, reaching a maximum grafting efficiency of 80% at a SDS amount of 3 wt.% and then it decreased. Surfactant could provide the monomer-swollen micelles favouring the diffusion of hydrophobic monomer onto the SiO_2 surface and the stability of PIP–SiO_2 nanocomposite resulted in an increase in grafting efficiency [33]. However, the grafting efficiency decreased at a high surfactant concentration due to the formation of ungrafted polyisoprene (free homopolymer) rather than the encapsulation of SiO_2 with PIP.

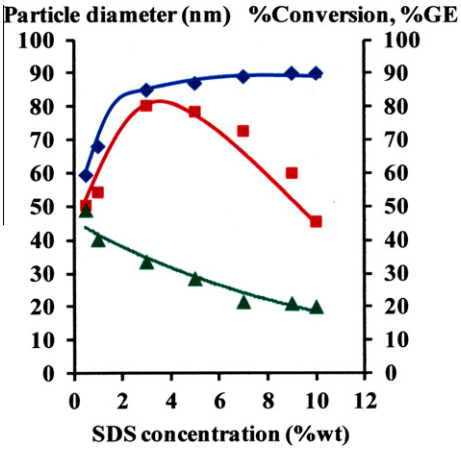


Fig. 4. Effect of SDS concentration on PIP–SiO_2; (▲) Particle diameter, (♦) %Conversion, (■) %GE. Condition: M/H_2O = 0.3, SPS = 1 wt.%, VTS–SiO_2 = 10 wt.% based on monomer.

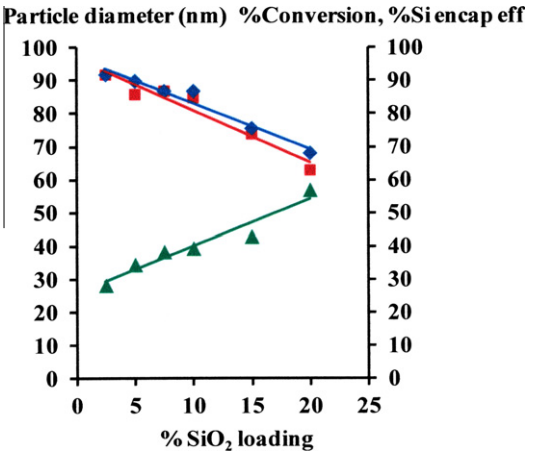


Fig. 5. Effect of silica loading on PIP–SiO_2; (▲) Particle diameter, (■) %Si encapsulation, (♦) %Conversion. Condition: M/H_2O = 0.3, SPS = 1 wt.%, SDS = 3 wt.% based on monomer.

The concentration of silica loading had a significant influence on particle size and encapsulation efficiency as shown in Fig. 5. When the silica level was increased from 5 to 20 wt.% based on monomer, the average particle size linearly increased from 29 nm to 57 nm and the silica encapsulation efficiency and conversion linearly decreased. This implies that at high silica loading, more aggregation of silica particles was produced and SiO_2 particles were not encapsulated, resulting in low encapsulation efficiency and large particle size. It is also probable that, the aggregation of silica particles tended to decrease the number of monomer-swollen micelles and particle stability, and therefore the final conversion was decreased. However, the conversion and silica encapsulation efficiency of PIP–VTS–SiO_2 at 5–20% silica loading were higher than those of PIP/untreated SiO_2 (conversion = 51%, Si encapsulation eff = 22%). Therefore, an appropriate silica loading of 10–15% could be added in the encapsulation for the PIP/SiO_2 nanoparticles with a high conversion (70–80%), small particle size (40 nm), and provide high encapsulation efficiency (75–80%) via this novel method.

Furthermore, the silica loading affected the decomposition temperature of PIP–SiO_2 nanoparticles as shown in Fig. 6. The thermograms of PIP and PIP–SiO_2 show a one-step polymer degradation and provide smooth weight loss curves, indicating the good compatibility between PIP and nano-SiO_2. Comparison with PIP-untreated SiO_2, the untreated nano-SiO_2 had a initial weight loss

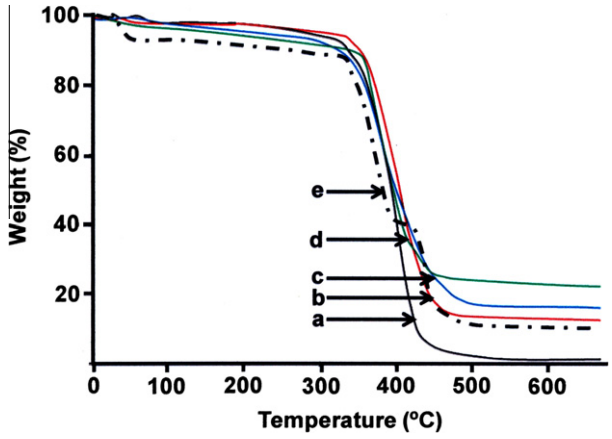


Fig. 6. Thermogram of (a) nanosized PIP, (b) PIP–VTS–SiO_2 (10 wt.%), (c) PIP–VTS–SiO_2 (15 wt.%,) (d) PIP–VTS–SiO_2 (20 wt.%), (e) PIP-untreated SiO_2 (10 wt.%).

below 100 °C which is related to the elimination of absorbed water from the silica surface [25] and a two-step weight loss at 390 and 430 °C was observed due to incompatibility between the silica and the polymer.

The SiO_2 content in the PIP/SiO_2 composite increased with increasing silica loading. From the thermogram, it is seen that the maximum decomposition temperature of PIP filled with VTS–SiO_2 at 0, 10, 15, 20 wt.% was 438, 455, 475, and 440 °C, respectively. This implies that the nano-SiO_2 could uniformly disperse in the PIP matrix resulting in the high thermal stability of the nanocomposite compared with PIP-untreated SiO_2. However, the decomposition temperature was decreased at high silica loading (20 wt.%) due to an agglomeration of nano-SiO_2 in the PIP matrix resulting in a reduction of the thermal resistance. Therefore, the nanosized PIP–SiO_2 at a silica loading of 10–15 wt.% led to high thermal stability.

Consequently, the optimum condition of PIP–VTS–SiO_2 synthesis was also applied for the synthesis of PIP–MPS–SiO_2 nanoparticles. It was found that at the same condition, the particle size of PIP–MPS–SiO_2 was 40 nm, monomer conversion was 84% and grafting efficiency was 74%. However, the type of silane coupling agents did not seem to have any significant effect on the synthesis of PIP–SiO_2 nanoparticles.

3.3. Particle size distribution of PIP–SiO_2 nanocomposite

The particle size distribution characterized by DLS was used to identify the dispersion of nano-SiO_2 in the PIP matrix. For PIP without SiO_2 particles, a particle size of 24 nm and a narrow size distribution of 0.02 were obtained as shown in Fig. 7a. For PIP/SiO_2 (untreated SiO_2 of 10 wt.%) (Fig. 7b), two-phase dispersion of nano-SiO_2 with average particle size of 10 nm and PIP domain with an average particle size of 22 nm was observed. It can be clearly seen that the untreated SiO_2 particles were not able to separate and stabilize due to the high polarity of their surface, resulting in heterogeneity and free polymer formation. Interestingly, the effective dispersion of silica in the PIP latex and a uniform size (average particle diameter of 35 nm) of the composite product were produced for PIP–VTS–SiO_2 (VTS–SiO_2 of 10 wt.%) as illustrated in Fig. 7c. The polydispersity index of PIP/unmodified SiO_2 (PDI = 0.16) was higher and the particle size distribution became boarder than that of PIP–VTS–SiO_2 (PDI = 0.02). This observation from the size distribution curves provides evidence for a uniform dispersion of PIP-modified SiO_2. Therefore, differential microemulsion polymerization of isoprene on modified SiO_2 could provide PIP–SiO_2 nanoparticles with a narrow size distribution resulting in homogeneous composite latex.

3.4. Morphology of PIP–SiO_2 nanocomposite

Morphology of PIP–SiO_2 nanocomposites characterized by TEM is illustrated in Fig. 8. For PIP/untreated SiO_2 (at 10 wt.%), silica particles as dark domains were aggregated inside the polyisoprene matrix as lighter domains (Fig. 8a). Free polyisoprene and agglomerated silica appeared; thus a heterogeneous structure of PIP–SiO_2 using untreated SiO_2 was observed due to interface debonding between the silica particle and polyisoprene matrix. For PIP–VTS–SiO_2 (at 10 wt.%) (Fig. 8b) and PIP–MPS–SiO_2 (at 10 wt.%) (Fig. 8c), these morphologies showed the encapsulation of dark SiO_2 cores with white polymeric layer shells around the core. Interestingly, the nanosized SiO_2 particles were well-dispersed inside the polymer latex as regular spheres and no agglomeration morphology of composite particles was predominantly observed. Therefore, nano-SiO_2 particles were encapsulated with a PIP shell, indicating that core/shell nanoparticles have been successfully produced from a differential microemulsion polymerization. The

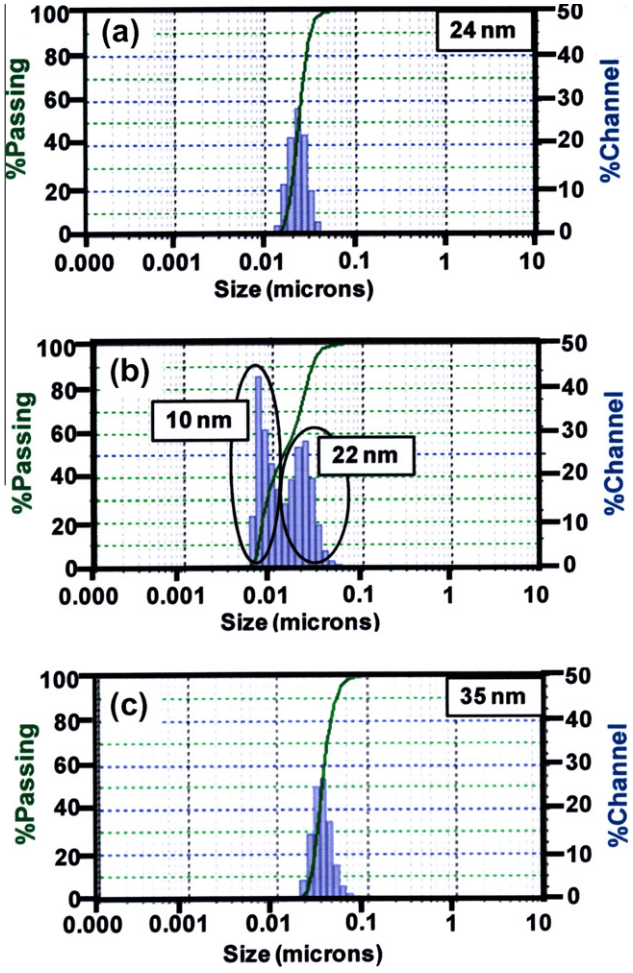


Fig. 7. Particle size distribution of composite sample (a) PIP, (b) PIP/untreated SiO_2 (10 wt.%) and (c) PIP–VTS–SiO_2 (10 wt.%).

average particle sizes of the composite particles as seen from the TEM photograph were about 30 nm and 40 nm respectively for PIP–VTS–SiO_2 and PIP–MPS–SiO_2 in good agreement with the DLS characterization.

3.5. Proposed mechanism for PIP–SiO_2 composite nanoparticle

A PIP–SiO_2 formation mechanism with controlled morphology of the core/shell is proposed in Scheme 2. The silica surface modification using organo-silane coupling agents was firstly carried out through hydrolysis and polycondensation to produce the carbon double bonds on the silica surface. For pretreatment, the polarity of the silica surface was decreased (Scheme 2a). Before polymerization, SPS initiator and SDS surfactant were dispersed in the aqueous silica dispersion. The interface between the oil phase and the aqueous phase could stabilize the colloidial particles by using SDS and the hydrophobic group of the surfactant tended to turn towards the oil phase and the hydrophilic group turned towards the aqueous phase used as the reaction medium (Scheme 2b). For a differential microemulsion polymerization, the system consisted of water, water-soluble initiator, nonaggregated surfactants, nonagglomerated silica and a very small amount of monomer droplets (isoprene). The initiator decomposed in an aqueous phase to generate free radicals which generated reactive monomer radicals on the silica surface for the initiation stage (Scheme 2c) and the primary monomer radicals were then polymerized to produce the oligomeric radicals resulting in chain propagation (Scheme 2d).

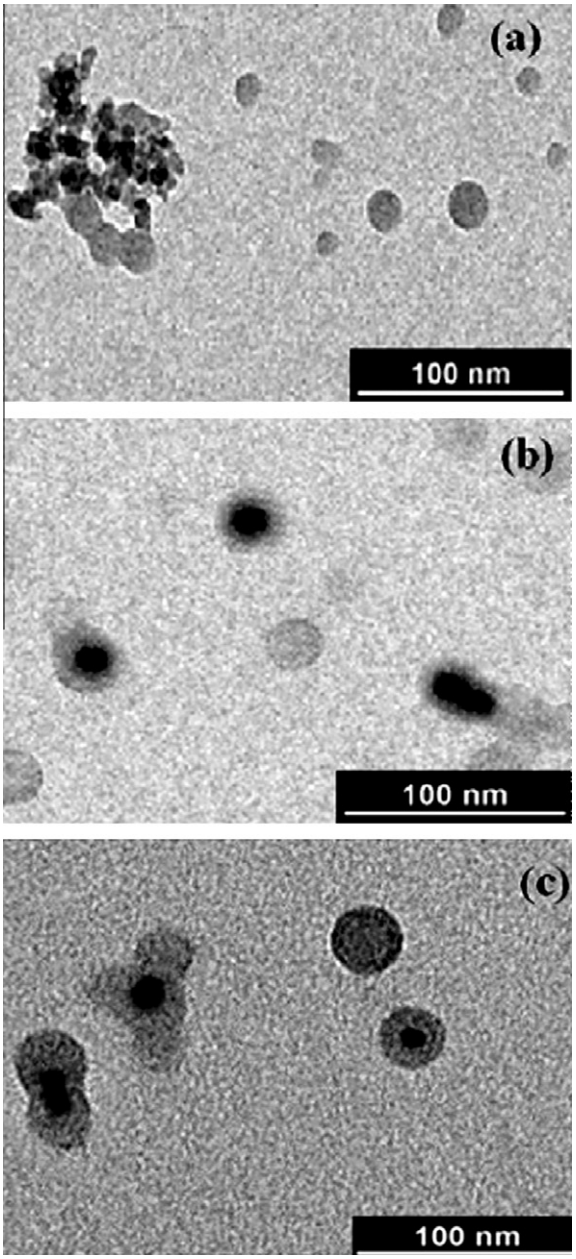
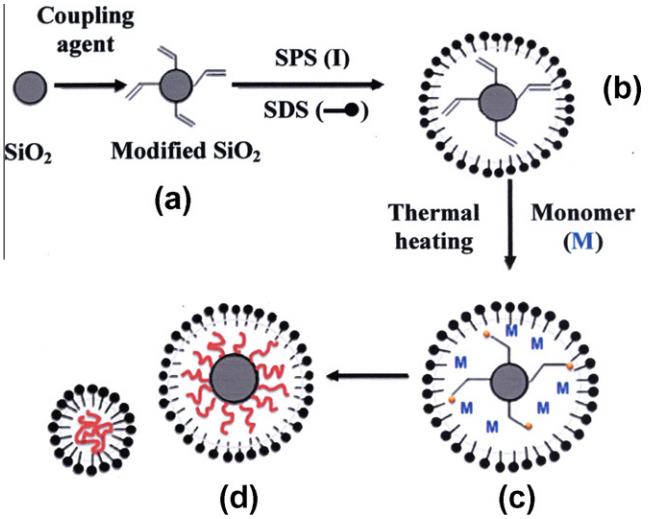


Fig. 8. TEM micrographs of (a) PIP/untreated SiO_2, (b) PIP–VTS–SiO_2, (c) PIP–MPS–SiO_2.

Therefore, the polymers could graft onto the silica surface through encapsulation to form silica core and polymer shell morphology.

3.6. Dynamic mechanical properties of NR/PIP–SiO_2 nanocomposite

Dynamic mechanical analysis is a powerful technique for the determination of a rubber modulus as a function of temperature and frequency. For an optimum condition, PIP–VTS–SiO_2 and PIP–MPS–SiO_2 at 10 wt.% of SiO_2 loading were selected to blend with NRL due to high monomer conversion and grafting efficiency. Modulus data in the form of the storage modulus (E′) of NR, NR filled with 20 wt.% of PIP–VTS–SiO_2 and 20 wt.% of PIP–MPS–SiO_2 are shown in Fig. 9a. The E′ values of NR and composites exhibited a plateau curve below the glass transition temperature (T_g) and decreased around the T_g. It is noted that the storage modulus of the unfilled NR (620 MPa) is much lower than that of the PIP–SiO_2



Scheme 2. The proposed formation mechanism of nanosize-SiO_2 encapsulated by nanosized PIP with core/shell morphology.

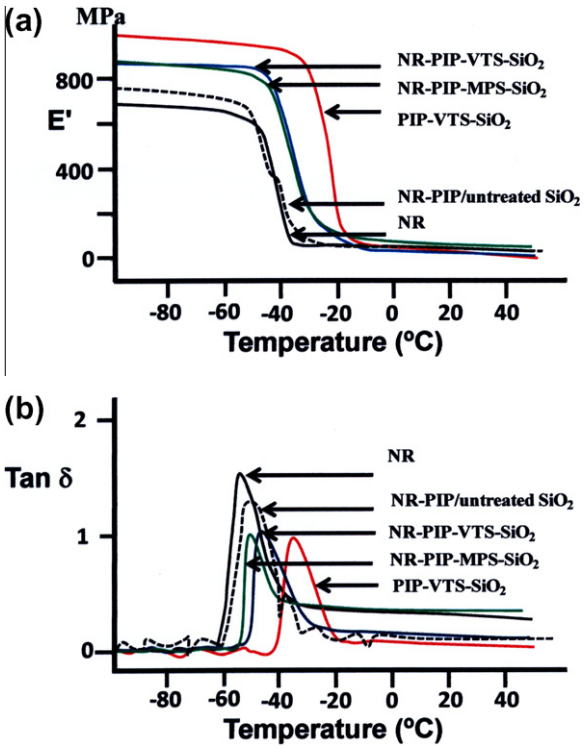


Fig. 9. Temperature dependence of (a) storage modulus (E') and (b) loss tangent ($\tan\delta$) for NR and its composites.

nanocomposite (915 MPa). In comparison with the NR composite, the E′ values of NR filled with PIP–VTS–SiO_2 (871 MPa) and NR filled with PIP–MPS–SiO_2 (860 MPa) were significantly higher than those of NR filled with 10 wt.% untreated SiO_2 (direct mixing) (710 MPa) due to a superior compatibility and a good dispersion of the nanofiller (PIP–SiO_2) within the NR matrix. Therefore, the improvement of the storage modulus could be achieved by the addition of monodispersed PIP–SiO_2 nanocomposite into the NR latex, resulting in the enhancement of strength and stiffness of NR.

The mechanical damping behavior on the temperature dependence for NR filled with 20 wt.% of PIP–VTS–SiO_2 and 20 wt.% of PIP/MPS–SiO_2 are given in Fig. 9b. It is clearly seen that the $\tan\delta$ is considerably decreased and the T_g is shifted to a higher temperature with the addition of PIP–SiO_2 into the NR matrix. NR and its composite had a single glass transition temperature and the T_g was −53 °C and −48 °C for unfilled NR and NR filled with PIP–VTS–SiO_2, respectively. The reduction of $\tan\delta$ which is determined from the ratio of the loss modulus (E″) to storage modulus (E′) implies the high storage modulus due to an increase in the homogeneity structure of the composite materials. Nevertheless, the addition of PIP–SiO_2 significantly affected the *Tg* value due to a lower mobility and flexibility of the polymer chains by the addition of rigid particles.

3.7. Mechanical properties of NR/PIP–SiO_2 nanocomposite

3.7.1. Mechanical properties of NR/PIP–SiO_2

From these results, high conversion, high%GE and narrow size distribution of PIP–SiO_2 were achieved at 10 wt.% of VTS–SiO_2 and 10 wt.% of MPS–SiO_2 amount. Therefore, PIP–SiO_2 at 10 wt.% of SiO_2 loading was selected to blend with NRL for mechanical testing. Mechanical properties of PIP–SiO_2 filled NR were investigated in terms of tensile strength, modulus, and elongation at break. The influence of the amount of PIP–VTS–SiO_2 and PIP/MPS–SiO_2 on the stress–strain behavior of NR nanocomposites was studied. Fig. 10a shows the stress–strain curves before ageing of unfilled NR and PIP–VTS–SiO_2 filled NR at ratio of NR to PIP–SiO_2 of 95/5, 90/10, 80/20, 70/30 (equivalent to 0.5%, 1%, 2%, and 3% SiO_2 content in all rubbers, respectively). It can be seen that the tensile stress of all PIP–VTS–SiO_2 filled NR were higher than that of unfilled NR,

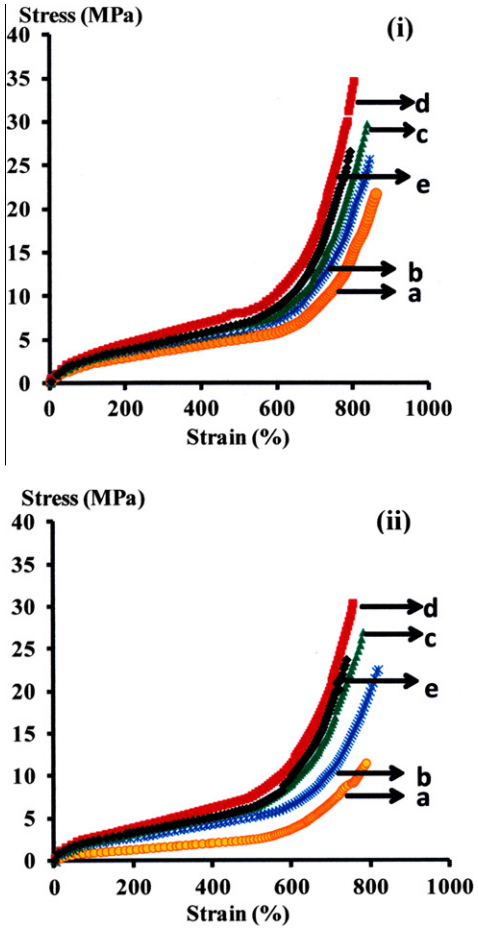


Fig. 10. Stress–strain curves of (i) PIP–VTS–SiO_2 filled NR before ageing; (a) NR, (b) 95:5 of NR:PIP–SiO_2, (c) 90:10 of NR:PIP–SiO_2, (d) 80:20 of NR:PIP–SiO_2, (e) 70:30 of NR:PIP–SiO_2 and (ii) PIP–VTS–SiO_2 filled NR after ageing.

indicating that PIP–VTS–SiO_2 filled NR exhibited a reinforcement effect. The tensile stress of NR nanocomposites gradually increased with an increase in PIP–VTS–SiO_2 amount from 5 to 20 wt.% and then decreased at an amount of 30 wt.%. For NR filled with PIP–MPS–SiO_2 nanoparticles as shown in Fig. 11a, the stress was increased with increasing the PIP–MPS–SiO_2 amount and the lowest tensile stress was observed for unfilled NR. The tensile stress of PIP–MPS–SiO_2 filled NR was slightly lower than that of PIP–VTS–SiO_2 filled NR. However the type of silane coupling agents for silica surface treatment did not show any significant effect on the physical properties of the NR/PIP–SiO_2 nanocomposite.

Tensile strength is an important property of rubber materials since it determines the limit of final stress in rubber applications. The tensile strength of the NR nanocomposite was significantly affected by the addition of PIP–VTS–SiO_2 and PIP–MPS–SiO_2. From Table 1, tensile strength is increased from 21.8 MPa (unfilled NR) to 34.6 MPa and modulus at 300% strains is increased from 3.9 MPa (unfilled NR) to 5.7 MPa with an increase in the amount of PIP–VTS–SiO_2 from 0 to 20 wt.%. For NR filled with 10 wt.% untreated SiO_2 (direct mixing), the tensile strength (28.8 MPa) and modulus at 300% strain (4.6 MPa) were lower than those of NR filled with PIP–VTS–SiO_2 and PIP–MPS–SiO_2. The result indicated that nanosized PIP–SiO_2 could increase the external force resistance resulting in a remarkable enhancement of tensile strength

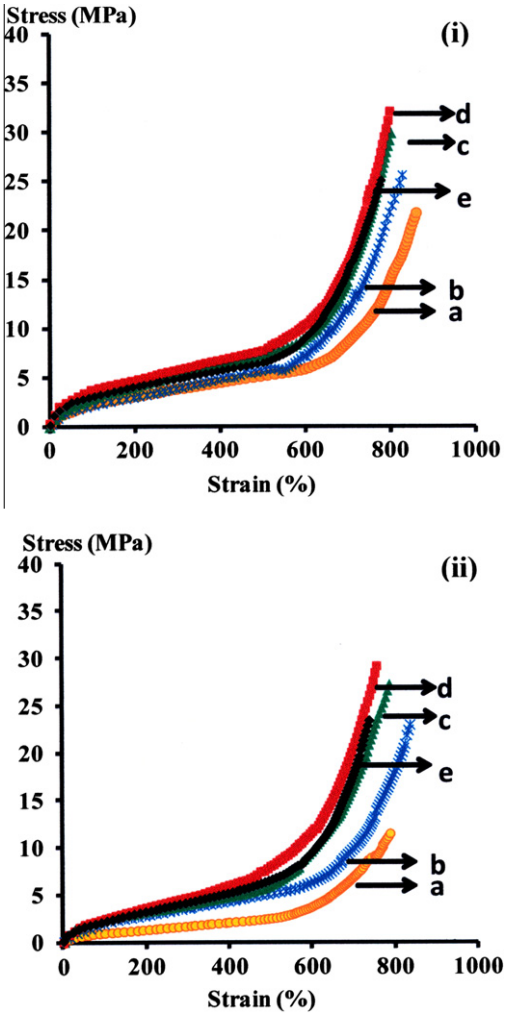


Fig. 11. Stress–strain curves of (i) PIP–MPS–SiO_2 filled NR before ageing; (a) NR, (b) 95:5 of NR:PIP–SiO_2, (c) 90:10 of NR:PIP–SiO_2, (d) 80:20 of NR:PIP–SiO_2, (e) 70:30 of NR:PIP–SiO_2 and (ii) PIP–MPS–SiO_2 filled NR after ageing.

and the incorporation of well-dispersed PIP–SiO_2 nanoparticles into the NR matrix and also led to the reduction of silica agglomeration. Furthermore, it can be explained in that the polyisoprene bonded on the silica surface tended to increase the compatibility within NR due to a major component of *cis*-1,4 polyisoprene in the NR structure. The tensile strength of PIP–MPS–SiO_2 filled NR had the same trend as PIP–VTS–SiO_2 filled NR. The mechanical properties of NR filled with 30% filler loading (PIP–VTS–SiO_2, PIP–MPS–SiO_2) were decreased due to the incompatibility between filler and rubber matrix and SiO_2 aggregation at high filler loading. A similar result was earlier reported for self-assembled natural rubber/silica nanocomposites [34]. The highest tensile strength of 32.1 MPa was achieved by adding PIP–MPS–SiO_2 in the amount of 20 wt.%. In comparison with unfilled NR, the tensile strength of PIP–VTS–SiO_2 filled NR and PIP–MPS–SiO_2 filled NR were at 1.6 and 1.5-fold of the unfilled NR, respectively. The PIP–VTS–SiO_2 and the PIP–MPS–SiO_2 amount of 20 wt.% is equivalent to 2% silica content in all rubbers. The tensile strength (34 MPa) and modulus at 300% strain (5.7 MPa) of NR nanocomposites were dramatically improved with an extremely low content of nano-SiO_2 (2 wt.%). From the previous work, a maximum tensile strength of 26.1 MPa and a modulus of 3.5 MPa of the NR filled with precipitated silica was obtained at a high silica loading of 30 wt.% [35].

Another remarkable improvement observed was for the modulus; the modulus at 300% strain was examined. From Table 1, a significant increase in the modulus of PIP–VTS–SiO_2 filled NR and PIP–MPS–SiO_2 filled NR were observed. The modulus at 300% strain was found to increase from 3.9 MPa (NR) to 5.7 MPa on increasing the PIP–VTS–SiO_2 amount from 0 to 20 wt.%. Moreover, the addition of PIP/MPS–SiO_2 over the range of 5–20% wt also resulted in an increase in modulus at 300% strain. The highest modulus at 300% strain was achieved at a ratio of NR to PIP–MPS–SiO_2 of 80:20. This implied that the improvement in the modulus of NR composites was due to the high interaction between the nanosized filler and the NR matrix [36].

For the elongation at break for unfilled NR and filled NR, it can be seen that unfilled NR exhibited the highest elongation at break (861%). NR rich compounds possessed the highest elongation at break due to NR crystallization which resulted upon stretching [29,37]. The %elongation at break linearly decreased with an increase in the nanofiller of PIP–VTS–SiO_2 and PIP–MPS–SiO_2 due to the hard and stiff nature of the silica particles. This phenomenon caused strain-reduced crystallization of NR dominating the elongation [34].

3.7.2. Thermal resistance of NR/PIP–SiO_2

To investigate the thermal resistance of PIP–SiO_2 filled NR composites, the influence of heat ageing on the mechanical properties of NR filled with PIP–VTS–SiO_2 and PIP–MPS–SiO_2 was studied as shown in Figs. 10b and 11b, respectively. The tensile stress of unfilled NR after heat ageing was greatly decreased over the range of strain studied. This indicated that the NR containing mainly the unsaturated carbon double bonds had poorer properties due to accelerated thermal ageing. On the other hand, the properties of NR filled with PIP–VTS–SiO_2 and PIP–MPS–SiO_2 did not change and retained high stress values after ageing. The percentage retention in tensile strength of unfilled NR (52.8%) was much lower than that of PIP–VTS–SiO_2 filled NR (89.9%) and PIP–MPS–SiO_2 filled NR (90.7%), respectively. This indicated that aged NR exhibited a reduction in NR crystallization and degradation of the polymer chains by high temperature treatment. For NR filled with 10 wt.% untreated SiO_2 (direct mixing), the% retention in tensile strength (71.1%) and modulus (50.0%) was lower than those of NR filled with PIP–VTS–SiO_2 and PIP–MPS–SiO_2, indicating a deterioration of properties with accelerated ageing [38]. The low mechanical properties of NR filled with 10 wt.% untreated SiO_2 (direct mixing)

Table 1
Mechanical properties of NR nanocomposites before and after ageing.

Rubber composites	NR/PIP–SiO_2[a]	SiO_2 content[b] (wt.%)	Tensile strength (MPa)		Modulus at 300% strain (MPa)		Elongation at break (%)	
			Before ageing	%Retention[c]	Before ageing	%Retention	Before ageing	%Retention
NR	–	–	21.8 (0.72)	52.8	3.9 (0.21)	45.0	861 (4.7)	91.6
NR/untreated SiO_2 [d]	90/10	10	28.8 (2.90)	71.1	4.6 (1.1)	50.0	799 (21.2)	81.3
NR/PIP–VTS–SiO_2	95/5	0.5	25.8 (0.66)	87.5	4.6 (0.02)	78.9	845 (12.8)	97.2
	90/10	1.0	29.7 (0.04)	90.5	4.9 (0.37)	84.0	839 (5.9)	93.3
	80/20	2.0	34.6 (0.74)	89.9	5.7 (0.70)	89.1	805 (15.3)	94.3
	70/30	3.0	29.6 (0.90)	80.0	4.8 (0.81)	87.9	794 (20.5)	93.4
NR/PIP–MPS–SiO_2	95/5	0.5	25.7 (0.47)	91.5	4.0 (0.36)	93.1	829 (1.4)	100.9
	90/10	1.0	29.9 (0.55)	93.4	5.2 (0.09)	75.2	803 (10.6)	98.1
	80/20	2.0	32.1 (0.81)	90.7	5.7 (0.85)	81.7	801 (19.9)	94.6
	70/30	3.0	25.0 (0.51)	93.2	4.7 (0.77)	88.0	780 (22.6)	94.7

Ageing conditions: 100 °C under air atmosphere for 24 h.
[a] NR/PIP–SiO_2, ratio of NR to PIP–SiO_2.
[b] Silica content based on total rubber.
[c] %Retention = (Properties after ageing/Properties before ageing) × 100.
[d] Direct mixing of untreated SiO_2 with NRL.

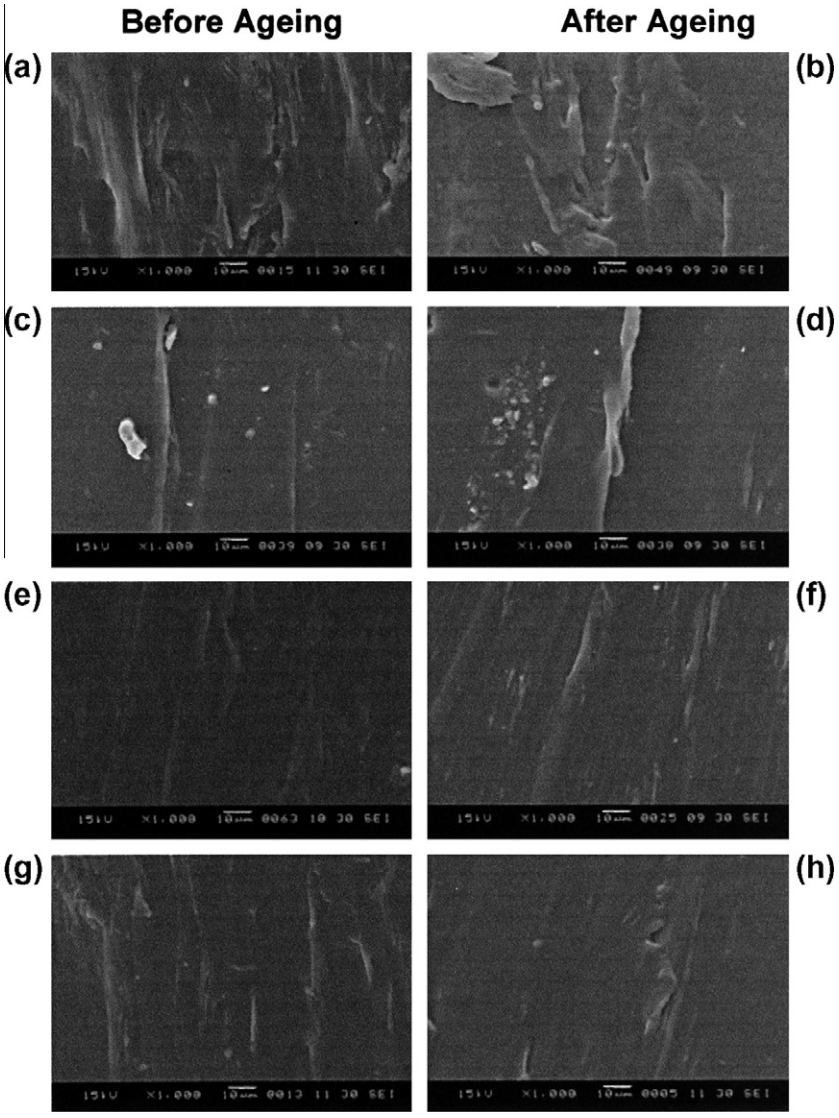


Fig. 12. SEM micrographs of samples before and after ageing (1000×). (a) NR, (b) aged NR, (c) NR filled with untreated SiO_2, (d) aged NR filled with untreated SiO_2, (e) NR filled with PIP–VTS–SiO_2, (f) aged NR filled with PIP–VTS–SiO_2, (g) NR filled with PIP–MPS–SiO_2, (h) aged NR filled with PIP–MPS–SiO_2.

are due to a poor interaction between SiO_2 and rubber matrix and silica agglomeration. For PIP-modified SiO_2 filled NR, PIP coveraged onto SiO_2 enhanced the improvement of the interaction between SiO_2 and NR. Similar results were previously observed for PMMA–SiO_2 filled PVC [39].

Another noticeable property observed was the modulus. The modulus at 300% strain after ageing of PIP–VTS–SiO_2 filled NR and PIP–MPS–SiO_2 filled NR remained constant (80–90%) as shown in Table 1. The percentage retention in modulus at 300% strain increased from 45 (NR) to 89% when the PIP–SiO_2 amount was increased from 0 to 20 wt.%. The lowest modulus retention of unfilled NR also showed poor heat resistance. This can be seen in that the enhancement in ageing resistance was dominated by the addition of well-dispersed PIP–SiO_2 nanoparticles into the rubber matrix. Similarly, the elongation at break of PIP–VTS–SiO_2 filled NR and PIP–MPS–SiO_2 filled NR after ageing linearly decreased with an increase in nanosized filler amount as shown in Table 1. The SiO_2 nanoparticles cause a reduction in the flexibility of the NR chain by restriction of the molecular chain slipping along the filler surface [31]. These results clearly demonstrated that the addition of PIP–VTS–SiO_2 and PIP–MPS–SiO_2 could increase the anti-ageing properties of NR products.

3.8. Surface morphology of NR/PIP–SiO_2 rubber nanocomposite

The surface morphology of unfilled NR, NR filled with PIP–VTS–SiO_2 at 20 wt.% (Si = 2%) and NR filled with PIP–MPS–SiO_2 at 20 wt.% (Si = 2%) before and after ageing was characterized by SEM as shown in Fig. 12. From Fig. 12a and b, it is seen that the surface of unfilled NR after ageing was smooth and exhibited a brittle appearance indicating the deformation failure in aged NR after high thermal ageing. Hence, unfilled NR had a lower heat resistance and inferior anti-ageing property. For the addition of nano-SiO_2 into the NR matrix, NR filled with unmodified SiO_2 had SiO_2 aggregates within the NR matrix, and thus phase separation and poor interaction were generated due to an incompatibility between the high polarity of the silanol groups on the silica surface and the nonpolar NR chain (Fig. 12c). For the addition of PIP–SiO_2 to NR, the surface of the NR nanocomposite exhibited the best compatibility between the nanofiller of PIP–VTS–SiO_2 and the NR matrix (Fig. 12e). This indicated that the high polarity of the silica surface was completely covered with nonpolar polyisoprene, and therefore the polyisoprene shell could improve the interfacial adhesion with *cis*-1,4 polyisoprene as a major component of NR. Similar results were also observed for PIP–MPS–SiO_2 filled NR (Fig. 12g). This implied that both PIP–VTS–SiO_2 and PIP–MPS–SiO_2 could increase the interaction bonding and the dispersion of the NR nanocomposite.

For surface morphology of NR composites after heat ageing, the adhesion between the unmodified SiO_2 and NR matrix was lower than that of modified SiO_2 due to aggregation of silica particles (Fig. 12d). For PIP–VTS–SiO_2 filled NR (Fig. 12f) and PIP–MPS–SiO_2 filled NR (Fig. 12h), the nanosilica particles were completely covered within the rubber matrix. This can be explained in that silica consists of hydrophilic groups of silanol on its surface while NR is composed of hydrophobic groups of *cis*-1,4 polyisoprene and thus, poor interaction and phase separation occurred. Nevertheless, the modified SiO_2 was strongly embedded with polyisoprene resulting in high interfacial adhesion and good compatibility with the NR matrix. After high thermal ageing, the SiO_2 aggregation and a phase separation was not clearly observed, implying that good miscibility of the filler in the rubber compound could be achieved through the encapsulation process and that the heat resistance of NR nanocomposites could be dramatically improved.

4. Conclusions

Well-dispersed polyisoprene–SiO_2 with a particle size of 35 nm was synthesized by a differential microemulsion polymerization technique in an aqueous system. The nano-SiO_2 was initially treated with organo-silane coupling agents of VTS or MPS. Isoprene was then polymerized, and therefore a nano-core (SiO_2) with a nano-shell thin layer (PIP) was achieved. From the study of process parameter effects, a high monomer conversion (87%), grafting efficiency (78%), and silica encapsulation (80%) was obtained at optimum reaction conditions at a monomer to water ratio of 0.3, at a low surfactant concentration of 3 wt.%, at an initiator concentration of 1 wt.% and at a SiO_2 loading of 10 wt.%. A small particle size of 35 nm with a narrow size distribution was also obtained. PIP–SiO_2 nanoparticles could be used as effective nanofiller for NR applications. For dynamic mechanical analysis, the storage modulus of NR filled with PIP–VTS–SiO_2 at 20 wt.% and NR filled with PIP–MPS–SiO_2 at 20 wt.% were found to be significantly higher than those of unfilled NR. Furthermore, $\tan\delta$ was decreased and the T_g was shifted to a higher temperature from -53 °C (unfilled NR) to -48 °C with the addition of PIP–VTS–SiO_2 at 20 wt.% into the NR matrix, indicating that the low dynamic loss modulus was due to an increase in the homogeneity structure of the composite materials. The higher *Tg* value was also due to a lower mobility of the polymer chains by the addition of rigid particles. For stress–strain behavior, the tensile stress of PIP–VTS–SiO_2 filled NR and PIP–MPS–SiO_2 filled NR was at 1.6 and 1.5-fold of unfilled NR, respectively due to miscibility and good adhesive interaction of the nanofiller (PIP–SiO_2) within the NR matrix, as confirmed by SEM. Interestingly, PIP–SiO_2 nanoparticles could improve the anti-ageing property of NR, by the high tensile strength retention of NR filled with PIP–SiO_2 (100%). Furthermore, the modulus retention increased from 45% to 89% with increasing PIP–SiO_2 amount from 0 to 20 wt.%. Therefore, the synthesis of the well-dispersed PIP–SiO_2 nanoparticles is of importance for rubber applications from an academic as well as an industrial point of view.

Acknowledgements

The authors gratefully acknowledge the support from the Thailand Research Fund (through the Royal Golden Jubilee Project), Graduate School, Chulalongkorn University, the Natural Sciences and Engineering Research Council of Canada (NSERC), the Thai Government Stimulus Package 2 (TKK2555) under the Project for Establishment of Comprehensive Center for Innovative Food, Health Products and Agriculture and the National Research University Project of CHE and Ratchadaphiseksomphot Endowment Fund (AM1024I).

References

[1] X.L. Xie, R.K.Y. Li, Q.X. Liu, Y.W. Mai, Structure-property relationships of in-situ PMMA modified nano-sized antimony trioxide filled poly(vinyl chloride) nanocomposites, Polymer 45 (2004) 2793–2802.
[2] R. Gangopadhyay, A. De, Conducting polymer nanocomposites, Chem. Mater. 12 (2000) 608–622.
[3] A. Laachachi, E. Leroy, M. Cochez, M. Ferriol, J.M. Lopez-Cuesta, Use of oxide nanoparticles and organoclays to improve thermal stability and fire retardancy of poly(methyl methacrylate), Polym. Degrad. Stab. 89 (2005) 344–352.
[4] E. Bourgeat-Lami, J. Lang, Encapsulation of inorganic particles by dispersion polymerization in polar media, J. Colloid Interf. Sci. 197 (1998) 293–308.
[5] A. Zhu, A. Cai, Z. Yu, W. Zhou, Film characterization of poly(styrene-butylacrylate-acrylic acid)–silica nanocomposite, Colloid Interf. Sci. 322 (2008) 51–58.
[6] S. Shang, L. Li, X. Yang, Y. Wei, Polymethylmethacrylate–carbon nanotubes composites prepared by microemulsion polymerization for gas sensor, Compos. Sci. Technol. 69 (2009) 1156–1159.
[7] F. Tiarks, K. Landfester, M. Antonietti, Silica nanoparticles as surfactants and fillers for latexes made by miniemulsion polymerization, Langmuir 17 (2001) 5775–5780.

[8] P. Xu, H. Wang, R. Tong, Q. Du, W. Zhong, Preparation and morphology of SiO_2/PMMA nanohybrids by microemulsion polymerization, Colloid Polym. Sci. 284 (2006) 755–762.
[9] G. Kickelbick, Concepts for the incorporation of inorganic building blocks into organic polymers on a nanoscale, Prog. Polym. Sci. 28 (2003) 83–114.
[10] G. He, Q. Pan, G.L. Rempel, Synthesis of poly(methyl methacrylate) Nanosize particles by differential microemulsion polymerization, Macromol. Rapid Commun. 24 (2003) 585–588.
[11] C. Norakankorn, Q. Pan, G.L. Rempel, S. Kiatkamjornwong, Synthesis of poly(methyl methacrylate) nanoparticle initiated by 2,2′-azoisobutyrontrile via differential microemulsion polymerization, Macromol. Rapid Commun. 28 (2007) 1029–1033.
[12] C. Norakankorn, Q. Pan, G.L. Rempel, S. Kiatkamjornwong, Synthesis of core/shell structure of glycidyl-functionalized poly(methyl methacrylate) latex particles via differential microemulsion polymerization, Eur. Polym. J. 45 (2009) 2977–2986.
[13] S. Chuayjuljit, A. Boonmahitthisud, Natural rubber nanocomposites using polystyrene-encapsulated nanosilica prepared by differential microemulsion polymerization, Appl. Surf. Sci. 256 (2010) 7211–7216.
[14] S. Chuayjuljit, W. Luecha, XSBR/NR rubber blends filled with polystyrene-encapsulated nanosilica prepared by in situ differential microemulsion polymerization, J. Elastomers Plast. 43 (2011) 407–427.
[15] G. He, Q. Pan, G.L. Rempel, Modeling of differential microemulsion polymerization for synthesizing nanosized poly(methyl methacrylate) particles, Ind. Eng. Chem. Res. 46 (2007) 1682–1689.
[16] N. Rattanasom, T. Saowapark, C. Deeprasertkul, Reinforcement of natural rubber with silica/carbon black hybrid filler, Polym. Test. 26 (2007) 369–377.
[17] P. Satrapan, A. Intasiri, V. Tangpasuthadol, S. Kiatkamjornwong, Effect of methyl methacrylate grafting and in situ silica particle formation on the morphology and mechanical properties of natural rubber composite films, Polym. Adv. Technol. 20 (2009) 473–486.
[18] Y. Urushihara, L. Li, J. Matsui, T. Nishino, In situ observation of filler displacement during tensile deformation of nanosilica-filled natural rubber using field-emission scanning electron microscope, Compos.: Part A 40 (2009) 232–234.
[19] P. Nontasorn, S. Chavadej, P. Rangsunvigit, J.H. O'Haver, S. Chaisirimahamorakot, N. Na-Ranong, Admicellar polymerization modified silica via a continuous stirred-tank reactor system: comparative properties of rubber compounding, Chem. Eng. J. 108 (2005) 213–218.
[20] M. Yooprasert, T. Pongprayoon, P. Suwanmala, K. Hemvichian, G. Tumcharern, Radiation-induced admicellar polymerization of isoprene on silica: effects of surfactant's chain length, Chem. Eng. J. 156 (2010) 193–199.
[21] D.M. Qi, Y.Z. Bao, Z.X. Weng, Z.M. Huang, Preparation of acrylate polymer/silica nanocomposite particles with high silica encapsulation efficiency via miniemulsion polymerization, Polymer 47 (2006) 4622–4629.
[22] J. Jang, H. Park, In situ sol–gel process of polystyrene/silica hybrid materials: effect of silane coupling agents, J. Appl. Polym. Sci. 85 (2002) 2074–2083.
[23] S.W. Zhang, S.X. Zhou, Y.M. Weng, L.M. Wu, Synthesis of SiO_2/polystyrene nanocomposite particles via miniemulsion polymerization, Langmuir 21 (2005) 2124–2128.
[24] I. Sondi, T.H. Fedynyshyn, R. Sinta, E. Matijevic, Encapsulation of nanosized silica by in situ polymerization of *tert*-butyl acrylate monomer, Langmuir 16 (2000) 9031–9034.
[25] Y. Sun, Z. Zhang, C.P. Wong, Study on mono-dispersed nano-size silica by surface modification for underfill applications, J. Colloid Interf. Sci. 292 (2005) 436–444.
[26] B.J. Jeon, H.J. Hah, S.M. Koo, Surface modification of silica particles with organoalkoxysilanes through two-step (acid-base) process in aqueous solution, J. Ceram. Process. Res. 3 (2002) 216–221.
[27] B. Suppaibulsuk, P. Prasassarakich, G.L. Rempel, Factorial design of nanosized polyisoprene synthesis via differential microemulsion polymerization, Polym. Adv. Technol. 21 (2010) 467–475.
[28] C. Norakankorn, Q. Pan, G.L. Rempel, S. Kiatkamjornwong, Synthesis of poly(methyl methacrylate) nanoparticles initiated by azobisisobutyronitrile using a differential microemulsion polymerization technique, J. Appl. Polym. Sci. 113 (2009) 375–382.
[29] B. Suppaibulsuk, P. Prasassarakich, G.L. Rempel, Synthesis of styrene-*g*-polyisoprene nanoparticles by emulsion polymerization and its effect on properties of polyisoprene composites, Polym. Adv. Technol. (2011), http://dx.doi.org/10.1002/pat.2069.
[30] S. Kiatkamjornwong, S. Apiwattanon, M. Rikukawa, N. Ogata, Super-fine particles of poly(styrene-*co*-methyl methacrylate) by dispersion polymerization, Colloids Surf. A 153 (1999) 229–240.
[31] J. Liu, Q. Pan, Synthesis of nanosized poly(ethyl acrylate) particles via differential emulsion polymerization, J. Appl. Polym. Sci. 102 (2006) 1609–1614.
[32] C. Norakankorn, Q. Pan, G.L. Rempel, S. Kiatkamjornwong, Synthesis of core/shell structure of glycidyl–functionalized poly(methyl methacrylate) latex nanoparticles via differential microemulsion polymerization, Eur. Polym. J. 45 (2009) 2977–2986.
[33] H. Xia, C. Zhang, Q. Wang, Study on ultrasonic induced encapsulating emulsion polymerization in the presence of nanoparticles, J. Appl. Polym. Sci. 80 (2001) 1130–1139.
[34] Z. Peng, L.X. Kong, S.D. Li, Y. Chen, M.F. Huang, Self-assembled natural rubber/silica nanocomposites: its preparation and characterization, Compos. Sci. Technol. 67 (2007) 3130–3139.
[35] H. Ismail, P.K. Freakley, I. Sutherland, E. Sheng, Effects of multifunctional additive on mechanical properties of silica filled natural rubber compound, Eur. Polym. J. 31 (1995) 1109–1117.
[36] S. Ono, Y. Kiuchi, J. Sawanobori, M. Ito, Structure development in silica-filled rubber composites, Polym. Int. 48 (1999) 1035–1041.
[37] S.H. Botros, Preparation and characteristics of NR/EPDM rubber blends, Polym. Plast. Technol. Eng. 41 (2002) 341–359.
[38] W. Arayapranee, G.L. Rempel, Morphology and mechanical properties of natural rubber and styrene-grafted natural rubber latex compounds, J. Appl. Polym. Sci. 109 (2008) 1395–1402.
[39] A. Zhu, Z. Shi, A. Cai, F. Zhao, T. Liao, Synthesis of core–shell PMMA–SiO_2 nanoparticles with suspension–dispersion–polymerization in an aqueous system and its effect on mechanical properties of PVC composites, Polym. Test. 27 (2008) 540–547.